

Juan Carlos Dominguez - Resume

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Projects

Data-Driven Insights for Early Breast Cancer Diagnosis | *Python, SQL, Tableau* Winter 2024

- Architected end-to-end SQL pipelines to efficiently process & aggregate large-scale diagnostic datasets, achieving a 40% reduction in data preparation time while ensuring data accuracy and scalability. Leveraged Python for advanced statistical modeling and in-depth feature correlation analysis, enabling precise identification of key diagnostic predictors.
- Uncovered transformative insights by analyzing diagnostic trends and patterns, leading to a 25% improvement in malignancy risk assessment accuracy. This work significantly contributes to refining diagnostic protocols and empowering clinicians to make more informed, data-backed decisions, improving patient care outcomes.

Comprehensive Academic Program Monitoring & Evaluation System | *Python, Flask, SQL, JavaScript* Fall 2024

- Engineered a relational database schema in MySQL to centralize academic data, supporting over 1000 records across degrees, courses, and instructors while maintaining 100% data integrity.
- Collaborated with a team to design and implement a scalable university database system by developing Flask-based RESTful APIs to connect the MySQL server database with a dynamic frontend interface, enabling real-time data management and improving system efficiency by 35%.

Scalable Machine Learning Model for Malaria Diagnosis | *Python, Sci-kit Learn, Machine Learning* Fall 2024

- Collaborated with a team of 4 to optimize the K-Nearest Neighbors classifier, achieving a significant accuracy improvement from 74% to 92% in identifying malaria-infected cells by implementing feature scaling, hyperparameter tuning, and robust data preprocessing pipelines.
- Streamlined team workflows by creating comprehensive documentation of data preprocessing pipelines and automating key steps using Python scripts, reducing data preparation time by 30%, allowing the team to focus on refining model performance.

NovaFit: Fullstack Fitness App with AI Assistance | *Python, JavaScript, Flask, AI Integration* Summer 2024

- Optimized database performance by implementing indexing, query optimization, and lazy loading strategies, achieving a 45% reduction in page load times and significantly enhancing the user experience.
- Designed and integrated an AI-powered chatbot using advanced natural language processing to provide personalized fitness recommendations and real-time assistance, leading to a 35% reduction in user drop-off rates.

Interactive & Dynamic Photography Portfolio Showcase | *HTML, CSS, JavaScript, React* Spring 2024

- Crafted a visually engaging, responsive website using advanced UX/UI principles to showcase diverse photographic styles, resulting in a 50% increase in user engagement and a significant boost in visitor interaction with portfolio content.
- Implemented Search Engine Optimization best practices, optimizing site architecture, metadata, and content, which led to a 30% increase in organic traffic. This improvement attracted a broader audience, enhancing the visibility and reach of the portfolio across search engines.

Work Experience

Frontend Software Developer Intern | *Running Logistics Inc.* Summer 2024

- Directed and led a team of 3 frontend developer interns in the planning and implementation of an effective login form, collaborating closely with the backend team to ensure seamless integration.
- Incorporated a responsive and user-friendly login interface that improved user experience and reduced form submission errors by 30%, facilitating smoother interactions for users.
- Consolidated with the backend team to analyze existing data and integrate keyword ranking algorithms, contributing to a 20% increase in the accuracy and relevance of resume matching, which enhanced the efficiency of the resume evaluation process.

Technical Skills

Programming Languages: HTML, CSS, JavaScript, Python, SQL, R

Technologies/Frameworks: React, Node.js, Git, REST APIs, Pandas, NumPy, TensorFlow, Flask

Developer Tools: GitHub, MySQL Workbench, Jupyter Notebook, Tableau, Power BI

Education

- M.S. Computer Science, *Southern Methodist University*
- B.S. Public Health, *University of California San Diego*

Expected Graduation: **May 2026**

Graduated: **June 2017**